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Lessons

Table of Contents

Confidence Levels

Notes

Bookmarks

Highlights

Using Inverse Operations

Toolbox

Using Inverse Operations

Think about this equation: $5 + x = 9$. Since you know that the sum of 5 and 4 is 9, you know the value of x must be 4. It is not difficult to see the solution to an equation with small whole numbers and one operation. But to find the solution to a complicated equation, you need a method.

An equation is solved when you find the value for the variable.

A solution can take either of these forms:

x	$=$	10	or	10	$=$	x
unknown		value		value		unknown

To solve an equation, you want to get the variable alone on one side of the equal sign. To do this, use **inverse operations**.

For example, if you have \$100 in savings and you spend \$20 (subtract), how do you get back to your original balance? You must deposit (add) \$20.

Addition and subtraction are inverse operations. If you subtract 20, you can undo the operation by adding 20.

The important thing to remember is that whatever you do to one side of an equation, you need to do the same to the other side to keep things equal. Think of an equation as a balance scale that must be balanced.

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Continue >>

Category
Lesson 3: Solving Equations

Related Questions:
[Practice questions taken related to this topic](#)

Example 1

[Hide](#)

Solve for y in the equation $y + 16 = 64$.
Think: You want to get y alone. The operation in the equation is addition. The inverse of addition is subtraction.
Do: Subtract 16 from both sides.
$$\begin{array}{r} y + 16 = 64 \\ -16 \quad -16 \\ \hline y = 48 \end{array}$$

The solution is $y = 48$.
Check: Substitute 48 for y in the equation.
$$\begin{array}{l} 48 + 16 = 64 \\ 64 = 64 \end{array}$$

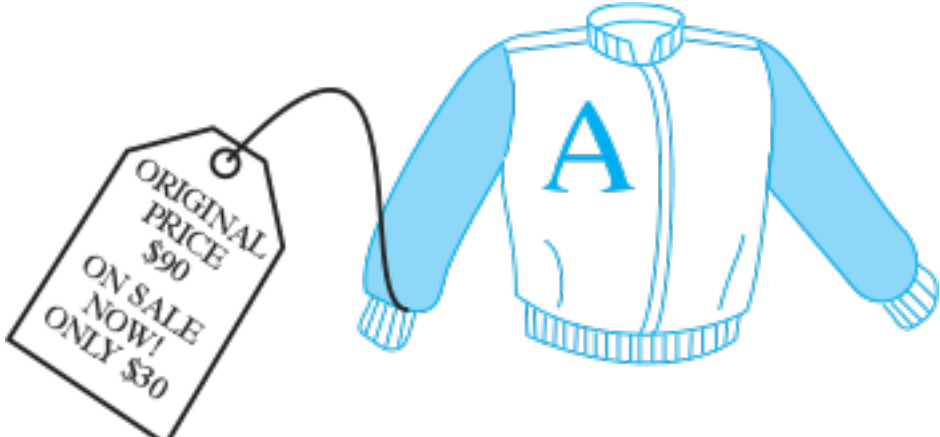
Example 2

[Hide](#)

Solve for n in the equation $3.25 = 2.4 + n - 9.2$.
Think: You want to get n alone.
First, combine like terms: $2.4 - 9.2 = -6.8$
The operation in the equation is subtraction.
The inverse operation is addition.
Do: Add 6.8 to both sides.
$$\begin{array}{r} 3.25 = n - 6.8 \\ +6.8 \quad +6.8 \\ \hline 10.05 = n \end{array}$$

The solution is $n = 10.05$.
Check: Substitute 10.05 for n in the equation.
$$\begin{array}{l} 3.25 = 2.4 + 10.05 - 9.2 \\ 3.25 = 3.25 \end{array}$$

Look at the price tag on the jacket in the drawing. The original price is 3 times the sale price. You can write two equations using this information. The inverse of multiplication is division.



$\$30 \times 3 = \90 $\$90 \div 3 = \30

Example 3

[Hide](#)

Solve for z in the equation $4z = 32.8$.
Think: You want to get z alone. The operation in the equation is multiplication. The inverse of multiplication is division.
Do: Divide both sides of the equation by 4.
(Remember, $4z \div 4 = 1z$ and $1z = z$.)
$$\frac{4z}{4} = \frac{32.8}{4}$$

$$z = 8.2$$

The solution is $z = 8.2$.
Check: Substitute 8.2 for z in the equation.
$$4(8.2) = 32.8, \text{ and } 32.8 = 32.8$$

Example 4

[Hide](#)

Solve for t in the equation $15 = -\frac{t}{9}$.
Think: You want to get t alone. The operation in the equation is division. The inverse operation is multiplication.
Do: Multiply both sides by -9 .
$$\begin{array}{l} 15(-9) = \left(-\frac{t}{9}\right)(-9) \\ -135 = t \end{array}$$

The solution is $-135 = t$.
Check: Substitute -135 for t in the equation.
$$15 = -\left(\frac{-135}{9}\right), \text{ and } 15 = 15$$

Some equations involve fractions. In the equation $\frac{3}{4}x = 12$, the variable x is multiplied by a fraction. Normally, you would divide to undo multiplication. With fractions, multiply by the **reciprocal**. The reciprocal is the number you multiply a fraction by to get a product of 1. The reciprocal of a fraction is found by switching the numerator and the denominator.

Example 5

[Hide](#)

The reciprocal of $\frac{1}{3}$ is 3 because $\frac{1}{3} \bullet \frac{3}{1} = \frac{3}{3} = 1$.

The reciprocal of $\frac{3}{4}$ is $\frac{4}{3}$ because $\frac{3}{4} \bullet \frac{4}{3} = \frac{12}{12} = 1$.

You can use reciprocals to solve problems.

Example 6

[Hide](#)

Solve for a in the equation $\frac{3}{5}a = 30$.
Step 1
Multiply both sides by the reciprocal of $\frac{3}{5}$. The reciprocal of $\frac{3}{5}$ is $\frac{5}{3}$.
$$\begin{array}{l} \frac{3}{5}a = 30 \\ \frac{5}{3} \bullet \frac{3}{5}a = \frac{5}{3} \bullet 30 \\ \frac{15}{15}a = \frac{150}{3} \\ a = 50 \end{array}$$

Step 2
Check.
$$\begin{array}{l} \frac{3}{5}a = 30 \\ \frac{3}{5} \bullet 50 \\ \frac{150}{5} = 30 \\ 30 = 30 \end{array}$$

Key Terms

- inverse operations
- reciprocal

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